

Worksheet · Section 3.17–3.18

Algorithmic Efficiency & Undecidable Problems



The Big Picture

Some problems are fast to solve, some are painfully slow, and some are impossible for any algorithm. Practice classifying efficiency, using heuristics, and spotting undecidable problems.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|--------------------------------------|---|
| _____ 1. Efficiency | A. Polynomial growth or better, like n or n^2 |
| _____ 2. Reasonable time | B. A problem with one specific input |
| _____ 3. Unreasonable time | C. How much time or memory an algorithm uses as the input grows |
| _____ 4. Heuristic | D. A shortcut that finds a good, not necessarily best, solution |
| _____ 5. Decision problem | E. Deciding whether any program will finish running or run forever |
| _____ 6. Optimization problem | F. Exponential or factorial growth, like 2^n or $n!$ |
| _____ 7. Decidable | G. A problem with a yes-or-no answer |
| _____ 8. Undecidable | H. No algorithm can solve every instance correctly |
| _____ 9. Halting problem | I. An algorithm can solve every instance correctly |
| _____ 10. Instance | J. A problem that asks for the best solution |

Part 2 · Fill It In

Use each word from the word bank once.

decision**exponential****undecidable****optimization****instances****Turing****heuristic****polynomial**

1. Algorithms that take n^2 steps run in _____ (reasonable) time.
2. Algorithms that take 2^n steps run in _____ (unreasonable) time.
3. A _____ finds a good-enough answer when the best one would take too long.
4. “Is 51 prime?” is a _____ problem.
5. “What’s the shortest route?” is an _____ problem.
6. A problem no algorithm can solve for every input is _____.
7. Alan _____ proved the Halting Problem is undecidable.
8. Some _____ of an undecidable problem can still be solved.

Name: _____

Class: _____

Date: _____

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Practice

Part 3 · Reasonable or Unreasonable?

Check the category for each number of steps (n = input size).

Item	Reasonable	Unreasonable
1. n		
2. n^2		
3. n^3		
4. 2^n		
5. $n!$		
6. $\log_2 n$		
7. $100n$		
8. 3^n		

Part 4 · Count the Steps

n	n^2	2^n
5	_____	_____
10	_____	_____
20	_____	_____

A REPEAT n TIMES loop inside another REPEAT n TIMES loop runs its inner body _____ times.If an n^2 algorithm takes 1 second for $n = 1,000$, it takes about _____ seconds for $n = 2,000$.

Part 5 · Classify the Problem

Problem	Decision or optimization?	Decidable?
Is this list sorted?	_____	_____
Find the cheapest set of flights for a trip	_____	_____
Will this arbitrary program ever stop?	_____	_____
Does the number 7 appear in this list?	_____	_____

Part 6 · Heuristic Thinking

A delivery driver has 40 stops. Explain why checking every possible route is impractical, describe a heuristic the driver could use, and name the trade-off.

**Think:**

Slow is not impossible! An unreasonable-time problem can be solved — it just takes too long for big inputs. An undecidable problem can't be solved for every input no matter how much time you have.

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. Which growth rate is unreasonable?

- A. n
- B. n^2
- C. 2^n
- D. $\log_2 n$

★★ On Level

2. An algorithm takes 1 second when $n = 10$ and 4 seconds when $n = 20$. Which best describes it?

- A. About n steps, reasonable
- B. Undecidable
- C. About n^2 steps, reasonable
- D. About 2^n steps, unreasonable

★★ On Level

3. Which problem is undecidable?

- A. Finding the largest number in a list
- B. Sorting a list
- C. Checking if a number is even
- D. Determining whether any given program halts

★★★ Exam Hard

4. A problem can be solved exactly, but only in exponential time. What is the best approach for very large inputs?

- A. Give up — it's undecidable
- B. Run it on one faster computer
- C. Use a heuristic to find a good approximate solution
- D. Use binary search

★★★ Exam Hard

5. Which TWO statements are true? (*Select two.*)

- A. No algorithm can solve all instances of an undecidable problem.
- B. A faster computer can make an undecidable problem decidable.
- C. Every optimization problem is undecidable.
- D. A heuristic may not find the optimal solution.

Part 8 · Written Response

Explain the difference between a problem that is “slow” and one that is “impossible” for computers. Give one example of each.
